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Combining a Direct and Indirect Training Approach for Cross-Domain Competences: The Case of the Course "Pedagogy for Psychotherapists"

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Abstract

In psychology programs, students should acquire both domain-specific knowledge and cross-domain competences important for later practice (e.g., multiple document literacy). Typically,

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such competences are trained *directly* in courses explicitly devoted to them or *indirectly* in courses on psychological topics that require them without systematically teaching and training them (e.g., when reading multiple texts about a topic). To exploit the advantages of both direct and indirect training approaches, we combined them in a new psychology course in which students were taught domain-specific knowledge on pedagogy and psychotherapy, and they were trained in the competence facets of (a) epistemic beliefs, (b) multiple document literacy, and (c) argumentative thinking. The direct training took a tried-and-tested example-based learning approach. A central element of the indirect training consisted of course assessments requiring the application of these three competences to the contents about pedagogy and psychotherapy. The combined training approach led to significant increases in declarative knowledge, advanced epistemic beliefs, and greater self-efficacy in implementing strategies relating to multiple document literacy and argumentative thinking. This approach can be adapted to accommodate different psychological content areas or different cross-domain competences.

Keywords

Cross-domain competences, example-based learning, direct and indirect training intervention

In psychology programs, as in other programs at university level, students should acquire domain-specific knowledge and cross-domain competences. For example, the German framework for qualifications in higher education (Kultusministerkonferenz; KMK, 2017) specifies that students should acquire discipline-specific knowledge, such as psychological theories, as well as more general, cross-domain competences, such as justifying decisions with theoretical knowledge, communicating, and cooperating with people outside their own field when solving problems, and taking into account diverging perspectives held by different stakeholders (KMK, 2017, p. 7). At the same time, such cross-domain competences should be grounded in and connected to domain-specific knowledge (e.g., justifying decisions with domain-specific theoretical knowledge; KMK, 2017).

However, it seems fair to say that the enhancement of students' cross-domain competences is often less than optimally addressed. Two typical ways can be distinguished. First, a study program may aim to foster a number of cross-domain competences (e.g., critical thinking, multiple document literacy), but most are not addressed explicitly by certain courses or lessons within courses. It is (implicitly) assumed that practicing such competences while focusing on discipline-specific issues (e.g., critically discussing studies on cognitive processes in a seminar) will foster these competences. Such an approach is an indirect training approach (e.g., Dignath & Büttner, 2018; Schuster et al., 2020): A course provides opportunities to practice competences or even demands them, but there is no direct teaching. While learners are expected to acquire a competence, the principles of said competence are not explained by the instructor, and difficulties of applying the competence are usually not addressed. Counting just on indirect training is usually regarded as suboptimal, as practicing competences without explicit teaching will not suffice to foster the deliberate, self-directed use of the competences (Cartier et al., 2010; Dignath & Büttner, 2018; Schuster et al., 2020).

Second, in some programs, students are explicitly taught cross-domain competences, taking a fully direct teaching approach. There are courses particularly designed to teach such competences, often offered by special centers for students from different study programs, by experts in that area, not regular instructors (e.g., centers for further qualification or student support). Hence, there is no close connection to the discipline-specific knowledge. While the principles of a skill are addressed directly, these are not applied to course-specific contents. In this case, it is questionable whether the learned competences will transfer when students later qualify in their own subject specialty program (e.g., Onstenk & Brown, 2002). This transfer problem is partly due to the fact that the direct approach is not combined with an indirect approach (Dignath & Büttner, 2018; Schuster et al., 2020).

Considering the shortcomings of direct and indirect training approaches, combining both approaches could prove effective for fostering cross-domain competences and their application and transfer to domain-specific knowledge. Previous interventions taking an example-based approach have shown that cross-domain competences can be fostered effectively (e.g., Hefter et al., 2014, 2015, 2018). We therefore developed a course in which cross-domain competences were trained both directly and indirectly in combination with domain-specific knowledge. In this report, we focus on the direct intervention components because the direct training can also be integrated into courses with other subject contents.

Training Background: The "Pedagogy for Psychotherapists" Course

In 2020, the German government adopted a new law ("Psychotherapeutengesetz") specifying a study program for psychology students that provides the option to obtain a license for psychotherapy ("Approbation"). Within this program, students complete coordinated Bachelor's and Master's degrees in Psychotherapy. At the Bachelor level, several courses are specified by the program as pre-requisites to qualify for the subsequent Master's degree. One of these is the course "Pedagogy for Psychotherapists," which prescribes four content fields: (a) education and personality formation ("Erziehung und Bildung"), (b) social and cultural factors in educational processes, (c) educational interventions and intervention settings, and (d) legal as well as family and social policy regulations influencing educational and psychological interventions. In addition, these contents should be related to the field of psychotherapy ("Pedagogy for *Psychotherapists*"). To reach the goal of integrating different knowledge types, two research groups at the Department of Psychology, University of Freiburg, cooperatively designed the course content and structure: the Educational and Developmental Psychology and the Clinical Psychology and Psychotherapy research groups. This course was designed for students at any stage in their Bachelor's degree, without needing prior knowledge about psychotherapy or any pre-requisite courses.

To integrate knowledge about the four content fields and about psychotherapy, we provided students with psychological case studies to be analyzed twice during the course. These case studies required integrating the different knowledge areas and, in particular, knowledge from the fields of education and psychotherapy. For example, students should decide on the best approach for an 18-year-old woman previously institutionalized because of suicidal tendencies and diagnosed with acute psychosis; but who wished to be discharged and needed help finding adequate housing and therapeutic support outside the hospital. The students needed to consider the perspectives of the patient, her psychotherapist, her separated parents, and her social worker to decide about which options to recommend to the patient. They needed to integrate their knowledge of different psychological theories, social policy regulations, and the influences of family-related processes in relation to that specific case study.

However, students find integrating knowledge from different areas quite challenging. First, to analyze their case studies, students have to integrate knowledge from the various documents they have studied during the course, which provide information about the four different content areas. Hence, it was beneficial if students had some multiple document literacy (MDL) for dealing with and integrating information from different sources (Saux et al., 2021; Stadtler & Bromme, 2013; Wineburg, 1991). Second, as these various types of knowledge or perspectives might not always suggest the same interpretations of the cases in question and the same therapeutic and educational actions, students should be able to consider and coordinate multiple perspectives when working on such case studies. This competence is often characterized as argumentative thinking (Iordanou et al., 2019; Kuhn, 1991). Third, advanced epistemic beliefs support the willingness to engage with multiple perspectives and multiple documents, and are an essential part of understanding how science works (Hefter et al., 2018; Kuhn et al., 2000). For example, it is important that students have understood (a) that knowledge needed to analyze complex case studies does not consist of a simple list of facts but of interconnected and not always entirely consistent information and (b) that there is often no one single correct perspective but rather different perspectives to consider and critically evaluate (e.g., Kuhn et al., 2000; Schommer, 1993; Stahl & Bromme, 2007). In a nutshell, for students' learning in the course "Pedagogy for Psychotherapists," it would help if students possessed the competences of MDL, argumentative thinking, and advanced epistemic beliefs.

Direct Training of Cross-Domain Competences: Example-Based Kick-Off-Interventions

Enhancing cross-domain competences is usually conceived as a long-term endeavor. Although we basically agree with this assumption (e.g., Kuhn, 2005), research has shown that kick-off training interventions based on the rationale of example-based learning can boost the initial development of cross-domain competences such as argumentative thinking (Hefter et al., 2014, 2015, 2018, 2019). These initial competences can also facilitate the acquisition of domain-specific knowledge (e.g., Iordanou et al., 2019).

Hefter et al. (2014, 2015, 2018, 2019) and Schworm and Renkl (2007) developed direct, example-based interventions to foster the development of cross-domain competences in a series of studies. Their interventions followed the same instructional principles (see Table 1). First, learners were given *explicit learning goals*. Then, the competence's key principles and cognitive steps

Table 1. Instructional Components Used in the Interventions by Hefter et al.

Instructional component	Example content from epistemic understanding intervention
Learning goals	Learning goal relating, for example, to knowing three levels of epistemic understanding
Theoretical introduction of key characteristics and consequences	Introduction of three levels of epistemic understanding (absolutism, multiplism, evaluativism)
Video-based examples, showing interaction between novice and expert, segmented	Novice showing lowest level of epistemic understanding, expert giving comments about characteristics and consequences of current level, novice reaching highest level at the end of interaction; video paused after each level modeled by novice
Self-explanation prompts (after each segment)	Prompt relating to level of epistemic understanding being shown in video
Self-regulated application phase	Video about two conflicting positions, followed by question to write argumentation and find own position

Note. Adapted from Hefter et al. (2014, p. 941).

were presented in a short *theoretical introduction* about the topic. This introduction aimed to enable learners to recognize characteristics and consequences of the competence shown in the worked examples. After the introduction, learners were given *video examples* showing the competence's application by a novice and an expert, where the expert made comments about the characteristics and consequences of the competence in question. The videos were *segmented* into three to four sections matching the components of the given competence, and learners were *prompted to self-explain* the principles shown in each section. In a subsequent self-regulated *application phase*, learners were asked to practice applying the competence without support, using content knowledge from a new example.

This approach was taken to foster various aspects of epistemic beliefs and argumentation skills including declarative knowledge, conceptual knowledge, and skill application (Hefter et al., 2014, 2015, 2018, 2019; Schworm & Renkl, 2007). Schworm and Renkl (2007) found that self-explaining examples using prompts that direct attention to underlying principles can serve to teach both declarative knowledge and the skill of argumentation. Based on this finding, Hefter et al. (2014) found that their training intervention, including the explicit learning goals, theoretical introduction, and self-regulated application phase, foster declarative knowledge and different components of argumentation skills, such as evaluative knowledge and argument quality. In a subsequent study, the authors found that the same approach could be taken to foster precursors of epistemic understanding, intellectual values, and conceptual knowledge about these two concepts (Hefter et al., 2015). In a combined intervention, Hefter et al. (2018) succeeded in fostering components of argumentation and of epistemic understanding in particular via an intervention that focused initially on epistemic issues, shifting later to focus on argumentation. Lastly, these authors compared various presentation modes for the worked examples, demonstrating similar learning outcomes following video-based, text-based, and graphic novel-based examples (Hefter et al., 2019). Together, their series of studies shows that this training approach successfully fosters cross-domain competences.

While this form of direct training is an effective means of supporting cross-domain competences, they were applied to content knowledge not connected to a specific domain relevant to learners. We expanded upon this training method by including three interrelated cross-domain competences in the same training module and by incorporating this training module in the new "Pedagogy for Psychotherapists" course, which provided contents relevant to the learners' study program. We will focus on the subjective learning outcomes in the form of students' self-efficacy for the MDL and argumentative thinking components, and on the development of more advanced epistemic beliefs.

Development of a Direct Training Module to be Integrated Into "Pedagogy for Psychotherapists"

Training Concept and Pilot Test

This training concept focused on three cross-domain competences. Epistemic beliefs were taught, as students should understand that topics covered in the "Pedagogy for Psychotherapists" course consist of interrelated concepts and not just a simple list of facts, and that this knowledge can change over time (e.g., on the basis of new evidence). Advanced epistemic beliefs also facilitate a fundamental understanding of science and the willingness to engage with different perspectives (Hefter et al., 2018; Kuhn et al., 2000). We taught MDL, as students had to learn to integrate different sources and different topics, as the course covered four different content areas of pedagogy as well as psychotherapy. MDL should also help students to view topics from different perspectives, and enable them to explore different options for future patients (Saux et al., 2021; Wineburg, 1991).

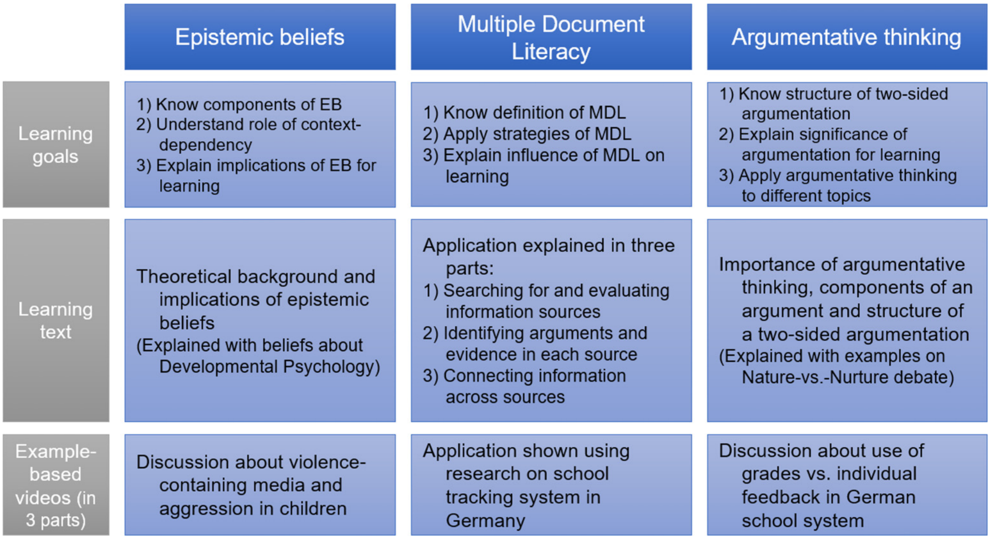

Figure 1. Overview of the training components.
Note. EB = epistemic beliefs, MDL = multiple document literacy.

Argumentative thinking was also taught, as students should be able to consider and coordinate different perspectives from different sources or in diverse content areas to help them analyze clinical case studies and draw well-founded conclusions (Iordanou et al., 2019; Kuhn, 1991).

Our training concept consisted of three parts relating to epistemic beliefs, MDL, and argumentative thinking, respectively (see Figure 1). Each part had three subsections, based on the instructional components used by Hefter et al.: first, the students were assigned explicit learning goals. Second, they were given an introductory learning text about the competence being taught. Third students received videos showing the competence's application using examples from domain-specific knowledge, which consisted of parts from the "Pedagogy for Psychotherapists" course. These videos were segmented according to each competence's main components to reduce potentially overwhelming cognitive load due to the transient nature of information from videos (Singh et al., 2012). Self-explanation prompts were provided after each segment to direct the learners' focus on principles (see Appendix A). Different topics were chosen from the "Pedagogy for Psychotherapists" course for the domain-specific knowledge used in the worked examples (e.g., how violence-containing media and aggression interrelate). Immediately before and after the training session, students were asked to write a short argumentation (500 words each, example excerpts in Appendix B) in a self-regulated application phase, on a topic from the "Pedagogy for Psychotherapists" course, after reading three short texts about the topic (i.e., gender segregation and the legal status of homeschooling, respectively).

The training concept was pilot-tested online using a sample of 171 adults recruited via flyers and social media websites, to examine how participants perceived the learning materials. The participants completed only the MDL part to make the training intervention shorter. Content knowledge from a different domain was used for the pilot test (i.e., developing a COVID-19 vaccine), which was thought to be more relevant to and easier for the chosen sample, to reduce the difficulty associated with the worked examples (Hilbert et al., 2008; Renkl, in press). The use of MDL was broken down into two main steps (evaluating sources, integrating information across sources). Participants were randomly allocated to one of the two groups. One condition using the MDL learning materials (including two subgroups with either continuous, $n = 57$, or diminishing support, $n = 55$), was compared to a condition focusing on presentation skills ($n = 59$) to see whether participants rated the newly constructed learning materials positively in comparison. We chose the comparison group's topic, as presentation skills are considered relevant for most professions and therefore interesting for the sample used. The learning materials were rated using four subjective ratings: experienced difficulty, situational interest, effort needed, and effort expended. As the two MDL conditions revealed no differences, they were combined for analysis.

As expected, in the pilot test we found no significant difference between the MDL ($M = 3.1$, $SD = 1.0$) and presentation skills ($M = 3.1$, $SD = 1.0$) groups for effort needed ($F(1,168) = 0.06$, $p = .809$). Nor was there any significant difference between the MDL ($M = 4.0$, $SD = 0.8$) and presentation skills ($M = 4.2$, $SD = 0.6$) groups in effort expended ($F(1,168) = 2.24$, $p = .137$) on the learning materials. However, we observed significant differences in situational interest ($F(1,168) = 17.14$, $p < .001$, $\eta_p^2 = .093$) and difficulty ($F(1,168) = 5.32$, $p = .022$, $\eta_p^2 = .031$). The intervention groups rated their interest significantly lower ($M = 3.2$, $SD = 0.9$) than the control group ($M = 3.8$, $SD = 0.8$) and difficulty significantly higher ($M = 2.8$, $SD = 0.9$) than the control group ($M = 2.5$, $SD = 0.9$). These findings suggest that the MDL learning materials were not perceived as positively as the presentation skills materials.

In response to these findings, we adjusted the learning materials by structuring the MDL content in three rather than two sections (evaluating sources, identifying arguments within sources, integrating information across sources) to help students understand and organize the learning content more effectively. As parts of the domain-specific knowledge from the "Pedagogy for Psychotherapists" course were used for the worked examples in the main study, this approach was thought to increase the interest in the training module.

Implementation

The training intervention was implemented during the first two weeks of the "Pedagogy for Psychotherapists" course (see Figure 2). All students participating in this course completed the

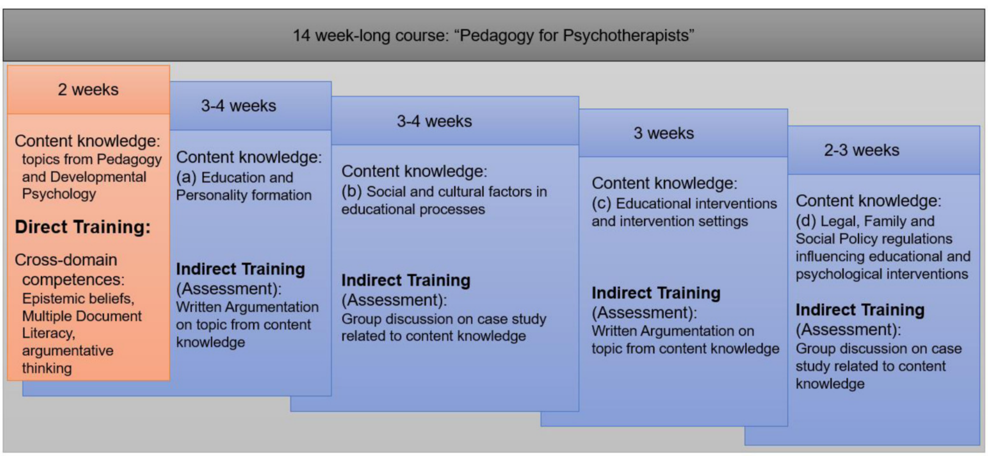

Figure 2. Overview of the "Pedagogy for Psychotherapists" course and training components.

training to prepare themselves for the rest of the course content. In this way, we combined direct and indirect training approaches.

The students were informed that the three cross-domain competences introduced in the training would help them understand the domain-specific knowledge from the different fields in the "Pedagogy for Psychotherapists" course. Students had to acquire the theoretical content knowledge in self-study of e-learning materials, which were divided into four sequences according to the four content fields. After each sequence, students applied their new knowledge to real-life questions and situations in one of the four course assessments. The cross-domain competences would also be important for these course assessments, for which students had to complete two written argumentations and two group discussions about case studies (see course syllabus in Appendix C).

Evaluation: Findings and Discussion

Forty-three undergraduate psychology students taking the "Pedagogy for Psychotherapists" course at a German university gave informed consent (33 female, 10 male, mean age = 25.7 years, SD = 5.8). The course was offered to students at different levels, but participants in this study were mostly in their third year of studies. Participants worked on tests and filled in questionnaires to assess their pre-training and post-training competence levels, which were provided immediately before and after the learning materials. All measures showed acceptable reliabilities (for an overview of measures and the corresponding Cronbach's α values, see Appendix D). Participants were asked to indicate their age, gender, how many years they had spent in education, and their high school Grade Point Average.

The students answered 15 true-or-false knowledge items to assess their declarative knowledge about each of the three cross-domain competences and the domain-specific knowledge from the "Pedagogy for Psychotherapists" course before completing the training (e.g., for MDL: "Your prior knowledge can help you decide whether an information source is credible."). After the training, students were given the same items a second time. They received one point for each correct answer, resulting in a scale from 0 to 15 points. The corresponding scale reliabilities were acceptable to good (all αs $\geq$ .63, see Appendix D).

Students also completed an epistemic beliefs questionnaire before and after the training. Epistemic beliefs were assessed using the short version of the German Connotative Aspects of Epistemological Beliefs (CAEB; Stahl & Bromme, 2007). This questionnaire uses 17 pairs of oppositional adjectives. Students rated how much scientific knowledge could be described by different poles of the adjective pairs on a scale from 0 to 7. The CAEB is commonly found to have a two-factor structure, with the factors Texture and Variability (Kienhues et al., 2008; Trevors et al., 2016). Texture refers to the beliefs about the structure and accuracy of knowledge, which range "from beliefs that knowledge is exact and structured to beliefs that it is unstructured and vague" (Stahl & Bromme, 2007, p. 778). Variability includes beliefs about the stability and dynamics of knowledge, ranging "from beliefs that knowledge is dynamic and flexible to beliefs that it is stable and inflexible" (Stahl & Bromme, 2007, p. 779). Texture is assessed using the sum of scores of nine of the 17 adjective pairs, variability using the sum of eight adjective pairs of the CAEB. Acceptable reliabilities have been reported in previous studies for both CAEB factors, with Cronbach's α values between .68 and .81 (Rosman et al., 2017; Stahl & Bromme, 2007), and we found similar reliability estimates (all αs $\geq$ .68). Finally, we assessed students' self-efficacy in using MDL and argumentative thinking via 12 questions (see Appendix E), where students were asked to indicate how confident they were in applying different strategies successfully, on an 11-point scale from "not at all confident" (0%) to "completely confident" (100%), as suggested

Table 2. Results of the Repeated-Measures ANOVAs Comparing Scores Before and After the Training.

	Before the training		After the training						
Knowledge tests	M	SD	M	SD	*F*	Model *df*	Error *df*	*p*	η_p^2
EB knowledge	7.13	3.47	11.13	2.56	50.76	1	23	<.001*	.688
MDL knowledge	10.09	3.61	12.50	2.43	19.18	1	33	<.001*	.368
AT knowledge	8.03	3.53	11.64	2.88	19.79	1	32	<.001*	.382
Domain-specific knowledge	8.24	3.07	11.50	3.09	23.98	1	33	<.001*	.421
Self-efficacy	63.56	20.79	75.86	14.98	21.83	1	36	<.001*	.377
EB: texture	23.18	7.15	25.53	7.39	4.95	1	39	.032*	.113
EB: variability	39.25	6.88	40.38	7.05	1.17	1	39	.286	.029

Note. ANOVA = analysis of variance; AT = argumentative thinking; EB = epistemic beliefs; MDL = multiple document literacy; SD = standard deviation; Self-efficacy = Self-efficacy for using MDL and argumentative thinking.
*p-value significant at $\alpha = .05$ level.

by Bandura (2006). The self-efficacy scores showed very good reliabilities (both αs ≥ .95; see Appendix D).

Students scored higher on all four declarative knowledge tests (about epistemic beliefs, MDL, argumentative thinking, and domain-specific knowledge) after the training than they had before, with very high effect sizes (Table 2). They also exhibited greater self-efficacy in applying MDL and argumentative thinking after the training than before (Table 2). Finally, students showed higher scores for the Texture factor of epistemic beliefs, but not for the Variability, after the intervention. Overall, these findings suggest that participants increased their knowledge about and willingness and confidence in applying skill components.

Overall Discussion

We developed a new Bachelor's level course on "Pedagogy for Psychotherapists" to meet one requirement in a combined Bachelor's and Master's degree study program qualifying students to become licensed to practice psychotherapy. In this course, students had to acquire both domain-specific knowledge from the field of psychology, and cross-domain competences that would help them process, weigh, and utilize information from different sources and perspectives. Against this background, we developed a brief direct training intervention to foster both the development of cross-domain competences and domain-specific knowledge. The three cross-domain competences were epistemic beliefs, MDL, and argumentative thinking, all of which can help students understand, process, and utilize information from the fields of study covered in the "Pedagogy for Psychotherapists" course (Bråten et al., 2011; Britt et al., 2014). We wanted to achieve these goals by fostering the development of the cross-domain competences using worked examples and incorporating elements of domain-specific knowledge from the course within these examples. We conceived of this direct training approach to enable the three trained competences to be applied to the content knowledge needed to pass the course. The direct training approach was, thus, combined with indirect training elements, in particular, with four course assignments in which the three cross-domain competences were required.

Our findings suggest that over the course of the training, students increased their declarative knowledge about the three cross-domain competences and about the domain-specific knowledge from the "Pedagogy for Psychotherapists" course. Students also showed more advanced epistemic beliefs and greater self-efficacy in using MDL and argumentative thinking after the training than before. The training can therefore be assumed as helpful in fostering the development of cross-domain competences and content knowledge taking a direct and an indirect training approach.

Students showed more advanced epistemic beliefs about the structure and accuracy of knowledge, as shown by the significant change in the Texture factor. However, we detected no increase in the Variability factor, that is, students did not seem to develop a more sophisticated view regarding the stability and dynamics of knowledge. These findings may be attributable to the topics chosen from the course content knowledge in the worked examples. Several of these topics, such as the influence of violence-containing media on aggression, have been debated for a long time and have not changed much over the years. This pattern may have created the impression that knowledge in these fields does not change substantially over time, reducing the influence of the training on epistemic beliefs. An updated training component about epistemic beliefs should focus more closely on the stability and dynamics of knowledge, and an exemplary topic should be chosen that does not suggest stability (e.g., the influence of homeschooling on learning during the Corona pandemic).

During the course, students reported a high workload associated with completing the training and working through the learning materials provided as part of the "Pedagogy for Psychotherapists" course (informal observation). For the future, we plan to redistribute the learning tasks and provide more support for students at the beginning of the semester. We plan to continue to provide this training without the assessment components (CAEB, self-efficacy questions, and declarative knowledge questions) to students taking this course in the future.

As all students had to acquire the domain-specific knowledge and the cross-domain competences to pass the course, we did not experimentally vary any factors, as that might have put some students at a disadvantage. However, to further develop our training approach, it would help to measure students' competences at the beginning and end of the "Pedagogy for Psychotherapists" course and then compare students who had undergone the direct training to those who only experienced the indirect training component (either the training at the beginning of the semester or the four course assignments that required the competences).

Our pilot test allowed us to adjust our learning materials before implementing them with our students. We utilized different topics as content knowledge for the examples; the topics were chosen to be interesting and relevant for the specific target group. For the pilot test during the first lockdown in 2020 with adults from the general population, we chose the development of the COVID-19 vaccine. For our course participants, we chose topics from the course they had to pass for their degree. As the topics were so different, they could have influenced learners' interest, and subsequently their learning outcomes. Generally speaking, the content knowledge of worked examples should not interfere with the learning outcomes of the competence being taught, as long as it does not require high levels of prior knowledge or cognitive load (Renkl et al., 2009). However, if learners are more interested, they may invest more effort into learning tasks, increasing the learning outcomes (Hidi & Renninger, 2006). Consequently, a measure of the interest should be included in future studies, to control for situational and individual interest in the content knowledge.

Our results show that students increased both their domain-specific knowledge and fundamental knowledge about the cross-domain competences simultaneously. It would be interesting to see whether having only one learning goal at a time, either the cross-domain competences or the domain-specific knowledge, would substantially influence learning outcomes. As our goal was to teach both components to students as part of the "Pedagogy for Psychotherapists" course, and students needed both components to pass this course, we did not include a control group with a single learning goal. Further research might investigate whether other options of integrating content knowledge and cross-domain competences would result in even better learning outcomes. For example, using concept maps to organize content knowledge from the marketing field was effective in teaching both content knowledge and the competence of concept mapping (Hilbert & Renkl, 2009). Working on argumentations about moral problems in genetics led to improved argumentation skills and content knowledge of genetics (Zohar & Nemet, 2002). Other studies show that using computers to learn about content knowledge can increase content knowledge, but computer use does not seem to improve media skills simultaneously (Schaumburg, 2006; Wecker et al., 2016). Therefore, integrating cross-domain competences and domain-specific knowledge may depend on the targeted competences and their fit to the content knowledge, or vice versa.

The direct training approach modeled here could be integrated into other courses in psychology programs, especially when they involve multiple perspectives. For example, developmental psychology courses often focus on different theories to explain the development of cognitive abilities in children, such as Piagetian, Vygotskian, and information processing perspectives. In clinical psychology courses, students often learn about various therapeutic approaches like those from cognitive-behavioral therapy, family-based psychotherapy, or psychodynamic approaches. In educational psychology courses, students may be exposed to cognitive-behavioral perspectives, cognitive-constructive approaches, and situated-learning approaches. In each of these examples, students are taught about different approaches or perspectives with evidence presented that supports or refutes each perspective, without completely rejecting any of them. Students need to understand how the approaches are related (i.e., similarities, how they complement each other, and contradictions), what advantages and disadvantages they have, and which perspective they want to take in a specific case. The three cross-domain competences trained in the current project would encourage such understanding. Training these competences should be more firmly implemented in university courses to reach important learning goals as specified by the German framework for qualifications in higher education (KMK, 2017).

Our training intervention seems to have been effective in supporting the development of three interconnected cross-domain competences that are appropriate for the contents of the "Pedagogy for Psychotherapists" course. In other courses, the training approach taken here could be adapted to cross-domain competences needed to teach specific content. For example, in one of our course assessments, students had to discuss a case study by adopting the roles of different stakeholders: patient, psychotherapist, mother, father, and social worker. Students had been trained to process different perspectives and weigh evidence when making decisions. However, drawing the same conclusion in group discussions may be facilitated by training students in discussion techniques and strategies for communicating and cooperating with others, especially with people from various fields or with different interests (e.g., Rummel et al., 2009). In other courses, such a competence could also be useful. For example, in courses on occupational or business psychology, students should gain this skill for discussions with employers, employees, and other stakeholders involved. Psychology students planning an academic career need to learn how to communicate with different cooperation partners, funding bodies, participants, and publishers. Psychology students who have already developed more fundamental competences, such as advanced epistemic beliefs and multiple document literacy may benefit from taking a combined direct and indirect training approach that incorporates communication and cooperation training. The training method described here could then be adapted to include the additional competences needed.

Conclusion

In conclusion, we have developed a combined direct and indirect training approach after which students showed advanced epistemic beliefs, higher self-efficacy in using MDL and argumentative thinking, and more domain-specific knowledge from the "Pedagogy for Psychotherapists" course. This training can take place at the start of the semester to prepare students to acquire psychological knowledge and for their later practice when they will have to make choices about options for patients. This training was implemented as part of a new "Pedagogy for Psychotherapists" course required in the new German Bachelor's degree in Psychotherapy qualifying students for a Master in Psychotherapy. We believe that such an intervention can also be adapted to other psychology courses to help students deal with multiple perspectives.

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Appendix A

Self-Explanation Prompts Provided During the Training and Examples of Typical Responses

Part	Prompt	Example of typical response
Self-explanation prompts for the epistemic beliefs part		
EB Part 1	Which levels of epistemic beliefs are being shown in this video and what characteristics identify them?	Anna (character A) is on the level of absolutism, she insists on her fixed view. Sofia (character B) is on the level of evaluativism, she combines several positions and possibilities and recognizes the versatility of the subject
	What can cause epistemic beliefs to change?	That one engages with the contents of a topic and consults different sources for it and looks at it critically
	Who do you find more convincing, and why?	The person on the left (character B) comes across as more convincing because their opinion is more complex and incorporates different points of view, which gives the feeling that they have done more research on the topic
EB Part 2	Which levels of epistemic beliefs are being shown now, and which characteristics indicate this?	The left person (character B) is on the level of evaluativism, since they include various studies to find a differentiated answer for themselves, but they are also aware that finding knowledge in this area is an unfinished process. The

(continued)

Appendix A. Continued.

Part	Prompt	Example of typical response
		person on the right (character A) is in absolutism because they expect a definitive and unquestionable answer from an authority, in this case the scientists
	How do the characters justify their claims and opinions?	Sofia (character B) refers to studies and experts, Anna (character A) focuses on her own opinion and what she knows in general in this area
	Which dimension of epistemic beliefs is noted in the last section?	Structure of knowledge
EB Part 3	Which dimensions of epistemic beliefs are mentioned in the videos and how was this shown?	The dimension of certainty is addressed by reflecting on whether and to what extent one can be certain of the state of knowledge or its development, or how one can contribute to it. The dimension of structure is also addressed when Sofia (character B) shows how research compiles facts, refers to previous knowledge, and thus integrates knowledge into a concept. Furthermore, at this point, reference is also made to sources and how they are to be evaluated and contrasted and how one can thus form and justify one's opinion. Thus, finally, the fourth and last dimension of epistemic beliefs is mentioned, namely the justification of knowledge
	What other distinction in epistemic beliefs were shown in this discussion?	Domain-specific, topic-specific, and general beliefs were addressed
	On what level of epistemic beliefs do both characters find themselves at the end of the discussion?	Evaluativism
Self-explanation prompts for the MDL part		
MDL Part 1	Why was the third article not considered very useful?	The multi-tiered nature of the school system was only briefly addressed and directly set aside, dismissed as irrelevant
	Why does the naming of an author lead to a higher evaluation of a source's reliability?	Since the motivation and expertise of the author can be included in the evaluation of reliability. If no author is indicated, it is also not possible to ask follow-up questions or to confront the author regarding his text (less responsibility for the author)

(continued)

Appendix A. Continued.

Part	Prompt	Example of typical response
	Why can the motivation of a scientific article's author be regarded as neutral or less relevant?	Since scientific articles must go through an objective review process before they can be published
MDL Part 2	Why is it harder to judge the claims and evidence of information sources regarding topics in which one has little knowledge?	Since the criteria of usefulness, verifiability and reliability cannot be judged well without prior knowledge. Thus, evidence is difficult to trace and the fit to the research question is also not clarifiable
	Which strategies do experts apply to evaluate information from texts, which are unnecessary in the current context?	Experts use contextualization to assess the credibility of sources. This means putting different sources into context, since authors are never completely objective
MDL Part 3	Which strategy was used to assess the Die Zeit article (German newspaper) that was not possible with the other sources?	The other articles could be mitigated as the statements of the article were compared to the others
	Why was it important to compare and contrast the Die Zeit article's information to the scientific article's?	Because they represent the results from different studies with different conditions
Self-explanation prompts for the argumentative thinking part		
AT Part 1	Which components of an argument are shown at the beginning of the discussion?	Claim, Reason, Warrant, Backing
	How is character A's position supported?	Using evidence that she has read a study, about how different teachers grade differently. She also supports her statement with her own assumption that bad grades cause bad feelings
	How does the argument's quality change, and why?	Evidence turns an opinion into a real argument and creates distinctness from other statements
AT Part 2	Which components of an argumentation have been used so far?	First theory + justification, other theories + justification, counterarguments + refutations
	Is character A's position stronger or weaker than it was at the beginning, and which aspect caused this change?	The position is stronger because she offers an alternative for the grading system that is already in use. Moreover, she finds counterarguments that do not speak against the new system, but rather states the lack of "competence of the teachers"
	How would you improve this argumentation?	By Anna (character A) refuting the counterarguments and combining the theories and counterarguments in a synthesis
AT Part 3	How did character B support character A's argumentation?	Sofia (character B) elicited the various components of a good argument from Anna (character A) with her questions and tested the strength of her point of view against alternative stances. Through clever questioning, Anna was able to show that she herself is extensively informed, as she was able to name and refute the counterarguments herself

(continued)

Appendix A. Continued.

Part	Prompt	Example of typical response
	Which components of an argumentation were especially important for strengthening character A's position?	I would say that each component has its justification, especially since they partly build on each other. But if I had to decide, I would say that the synthesis of arguments and theories is the most important part, because it concludes everything and contains the final opinion
	What effect does character A's last claim have?	That Sofia (character B) is convinced of Anna's (character A) position

Note. AT = argumentative thinking; EB = epistemic beliefs; MDL = multiple document literacy.

Appendix B

Example Excerpts From the Argumentations

The following excerpts (180–250 words) have been taken from longer argumentations (approx. 400–800 words) to show the quality of argumentations before and after the intervention took place.

Example 1: Argumentation About Gender Segregation in Schools (Before Intervention)

"(…) A third perspective highlights the complexity of this discussion by pointing to a 'larger' study of its own, which concluded that the previous contradictory findings were insufficient due to weak empirical evidence or partly wrong interpretation of the results (Halpern et al. 2011). An explanation of possible selection effects of the other studies underscores their own evidence. Additionally, the publication in Science is highlighted to give more weight to this argument. German researchers are probably sceptical about these results. This attitude is supported by more conclusive studies, which show certain advantages of separation (Kessels & Hannover, 2010). Another meta-analysis (Pahlke et al., 2014), and the most recent source to date, shows that well-controlled studies conclude that there is no difference between student achievement and interest. Finally, Viewpoint 3 would like to see an individualized opportunity to choose school orientation. All three views have drawn on various studies to support their argument. In fact, in the context of education, the emphasis was mainly on the perfect development and the possible achievement of the person. Personally, I find many social and societal arguments and evidence favoring segregation lacking here. Stereotypical ways of thinking still prevail in people's minds today. Not infrequently, there are accusations in disputes related to gender and would, in my opinion, by a gender separation in the young years still promote. Children should experience the freedom and diversity of humanity and must learn early that there is 'other' than an artificially created homogeneity. It is a chance and a game of hormones to deal with the opposite sex. (…)"

Example 2: Argumentation About Gender Segregation in Schools (Before Intervention)

"Gender segregation has advantages and disadvantages, as listed in the three essays I used as sources. The arguments for gender separation emphasize observed differences in boys and girls, for example, that boys need more time to learn to read and girls have more difficulty in math. A gender-segregated class could provide specific training for each weakness. Some studies have shown that gender separation is effective. Students in such classes have been shown to have a better social self-concept, a better appreciation of their abilities, and higher expectations for their future careers. These are all positive aspects, but a meta-analysis of studies that controlled for other variables shows no significant difference between performance in gender-segregated schools and integrated schools. There are many possible variables that can skew the data. Most gender-segregated schools are private schools that provide more resources for students. It is important to control the studies so that the independent variable of gender segregation is truly looked at and isolated. A meta-analysis of randomized control trials (RCTs) is the gold standard in empirical research. Evidence from meta-analysis is not the only argument against the need for sex segregation. Consideration must also be given to gender atypical students who would be left out entirely if gender segregation were used. (…)".

Example 3: Argumentation About the Legalization of Homeschooling (After Intervention)

"(…) The argument is also often made that children who are home schooled are more likely to pursue their own interests and also show more learning progress (Cheng et al., 2016). Again, however, the selection of study participants should be considered more closely. The sample could be children from privileged backgrounds. For example, children who participate in less structured homeschooling show significantly worse performance than children who attended school. Thus, heterogeneity in the quality of homeschooling should also be included in the consideration of studies. In school education, on the other hand, a homogeneous type of education is guaranteed by the extensive training of teachers. Finally, social consequences in particular play an essential role in homeschooling. It is difficult to qualitatively and objectively assess the socialization of homeschooled children. But interaction with peers is important for children's development and is significantly reduced by homeschooling. It has even been shown that children who were homeschooled severely lack social skills and opportunities to make friends (Neumann, 2020). It is clearly evident that homeschooling in Germany should not be solely in the hands of parents."

Example 4: Argumentation About the Legalization of Homeschooling (After Intervention)

"(…) Furthermore, it should be critically considered to what extent parents are competent to support their children with difficult problems or topics. Schooling offers children views on certain topics from different (objective) perspectives, which is very often not the case in homeschooling. According to Reich (2005), children learn less self-determination and are thus ill-prepared for the debates that occur in a diverse nation in many areas. Finally, it must be mentioned that homeschooling brings with it many changes in the family's routine and, as a result, parents may have less time for other obligations and activities. For this reason, it is especially important for each family to consider the advantages and disadvantages of homeschooling. A possible solution would be to combine these two systems and give the children, besides homeschooling, also the possibility to participate in different activities at school. Another problem with this topic is that it is particularly difficult to draw firm conclusions because there are not enough objective studies and evidence about the effects of homeschooling. Therefore, it would be useful to conduct several representative studies before deciding whether to allow certain forms of homeschooling."

Appendix C

Course Syllabus (Translated From German)

Course: "Pedagogy for Psychotherapists". In this qualifying module, theories and evidence about education and personality formation, learning difficulties and pedagogical interventions, as well as legal, family and social policy-related aspects will be addressed. Furthermore, cultural and social influences on education and personality formation will be considered. Students will acquire the theoretical knowledge contents in self-study using e-learning materials, with support from lecturers. This knowledge will then be discussed and put into practice using case studies during three in-person teaching events.

Pre-requisites: Students enrolled in Bachelor of Psychology degree.

Maximum number of students: 60.

Assessment: Independent study using e-learning materials divided into four sequences based on the four content fields (education and personality formation, social and cultural factors in educational processes, educational interventions and intervention settings, and legal as well as family and social policy regulations influencing educational and psychological interventions); active participation in three in-person teaching events; two written argumentations about topics from two content fields.

Examination: Take-home exam.

Materials: e-learning materials provided online.

Appendix D

Measures Used and Their Reliabilities

Concept	Measure/questionnaire	Reliability (Cronbach's alpha)
Epistemic beliefs	CAEB (Stahl & Bromme, 2007)	
Pre-test		
Texture		.78
Variability		.68
Post-test		
Texture		.80
Variability		.81
Prior knowledge	15 items each	
Epistemic beliefs		.75
MDL		.63
Argumentative thinking		.72
Domain-specific knowledge		.77

(continued)

Appendix A. Continued.

Concept	Measure/questionnaire	Reliability (Cronbach's alpha)
Declarative knowledge	15 items each	
post-test		
Epistemic beliefs		.75
MDL		.85
Argumentative thinking		.86
Domain-specific knowledge		.88
Self-efficacy	12 questions	
Pre-test		.95
Post-test		.96

Note. CAEB = Connotative Aspects of Epistemological Beliefs; MDL = multiple document literacy.

Appendix E

Questions Used to Assess Self-Efficacy in Using MDL and Argumentative Thinking

Q1: I can explain strategies that can be used for evaluating the usefulness, reliability, and verifiability of information sources.

Q2: I can use strategies to evaluate the usefulness, reliability, and verifiability of information sources.

Q3: I can explain how to connect information between different information sources.

Q4: I can use strategies to connect the information between different information sources.

Q5: After reading multiple texts, I can summarize the contents of these sources.

Q6: After reading multiple texts, I can represent the contents of these sources in relation to each other.

Q7: I can explain which components constitute an argument.

Q8: I can formulate a convincing argument with all its components.

Q9: I can explain which criteria have to be fulfilled for a valid claim.

Q10: I can formulate valid claims.

Q11: I can explain how a meaningful argumentation is constructed.

Q12: I can construct a meaningful argumentation.